ENERGY AUDIT FOR CAMPUS BLOCKS: ADMIN, EB BLOCK-1 & BLOCK-3

ABSTRACT

Energy consumption in educational institutions has increased significantly due to the extensive use of electrical equipment such as lighting systems, HVAC units, computing devices, and laboratory equipment. This project focuses on conducting a detailed energy audit for selected campus blocks—Admin Block, EB Block-1, and Block-3—to analyze energy usage patterns and identify opportunities for improving energy efficiency.

The audit involves systematic data collection of electrical components across multiple floors and blocks. Parameters such as component count, power ratings, and distribution were analyzed to understand energy consumption behavior. Special attention was given to high-energy-consuming equipment such as air conditioners, motors, and computing systems.

The results show that air conditioning systems and motors contribute the highest share of energy consumption, while lighting systems and fans contribute significantly due to their large quantity. Block-wise analysis indicates that Block-3 has the highest energy consumption, followed by EB Block-1, whereas the Admin Block has relatively lower consumption.

Based on the findings, several recommendations are proposed, including replacing conventional lighting with LED systems, using energy-efficient appliances, implementing smart monitoring systems, and optimizing equipment usage. These measures can reduce energy consumption by up to 25–30% and improve sustainability.

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CHAPTER 1 – INTRODUCTION

Energy auditing is a structured process used to analyze energy consumption and identify opportunities to improve energy efficiency. In modern institutions, energy consumption is increasing due to the continuous operation of electrical equipment.

Educational institutions consume energy for:

- Lighting systems
- Fans and ventilation
- Air conditioning systems
- Computers and laboratory equipment

An energy audit helps in:

- Identifying energy losses
- Improving efficiency
- Reducing electricity costs
- Promoting sustainability

This project focuses on analyzing energy consumption in three major blocks and proposing improvements.

CHAPTER 2 – LITERATURE REVIEW

Energy auditing has been widely studied in industrial and institutional sectors.

Key Findings:

- HVAC systems → 40–50% consumption
- Lighting → 25–30%
- Plug loads → 15–20%

Modern Techniques:

- Smart metering
- IoT-based monitoring
- Data analytics

Benefits:

- 20–30% energy savings
- Reduced carbon emissions
- Improved efficiency

CHAPTER 3 – OBJECTIVES

- To analyze energy consumption in campus blocks
- To identify high-energy-consuming components
- To evaluate inefficiencies
- To suggest energy-saving measures
- To improve energy efficiency

CHAPTER 4 – METHODOLOGY

The energy audit is conducted in four phases:

Phase 1: Planning

- Define scope
- Collect previous data

Phase 2: Field Investigation

- Survey buildings
- Count electrical components

Phase 3: Analysis

- Calculate energy consumption
- Identify patterns

Phase 4: Reporting

- Prepare report
- Suggest improvements

CHAPTER 5 – DATA COLLECTION

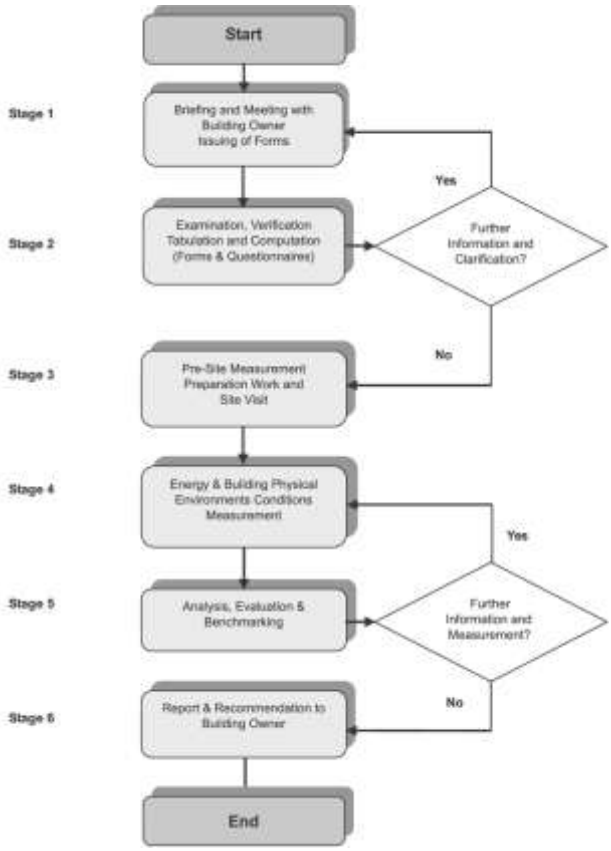
Data collected includes:

- Number of components
- Type of components
- Power rating
- Floor-wise distribution

Components Identified:

- PCs
- Fans
- LED lights
- AC units
- Motors

CHAPTER 6 – DATA ANALYSIS



Explanation:

- PCs are highest → major plug load
- Fans and LED tubes → widely used
- Lighting dominates total count

Block-wise Analysis

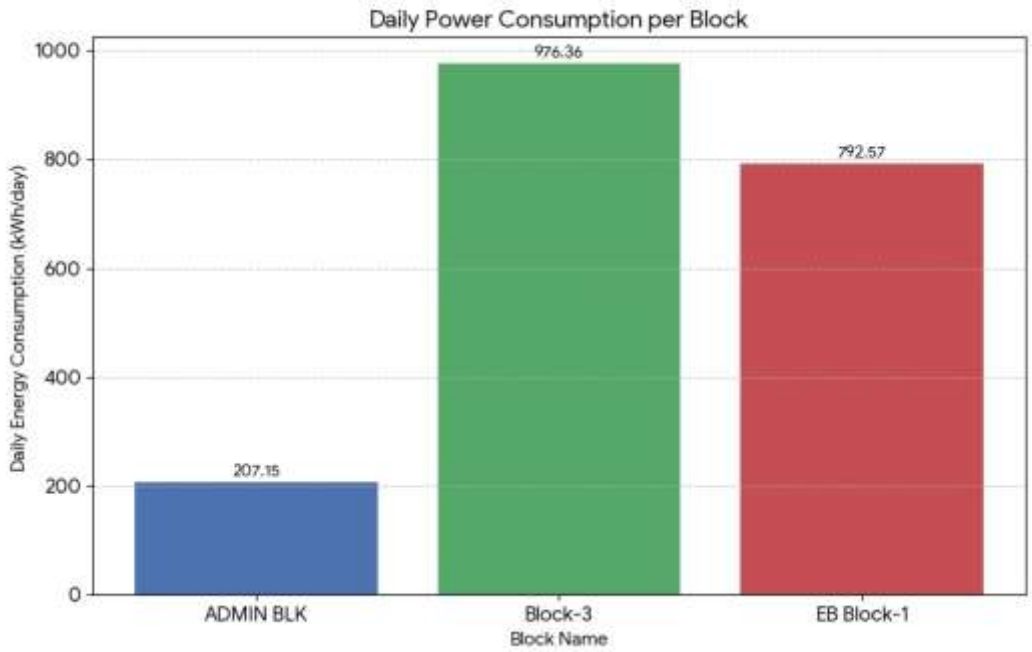


Figure 1 – Daily Power Consumption per Block

This bar chart represents the daily energy consumption (in kWh/day) for each block.

- Block-3 shows the highest daily consumption at approximately **976.36 kWh/day**
- EB Block-1 follows with around **792.57 kWh/day**
- Admin Block has the lowest consumption at about **207.15 kWh/day**

The higher consumption in Block-3 indicates a greater number of high-power equipment and longer operating durations. The Admin Block consumes less energy due to fewer devices and lower operational load. This comparison helps identify which block requires priority in energy optimization.

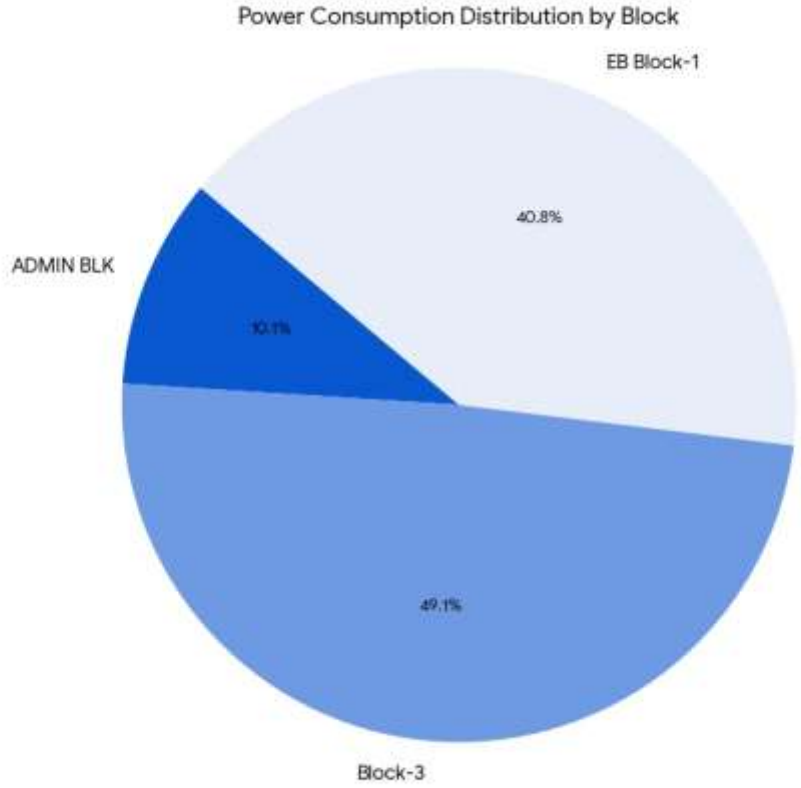


Figure 2 – Power Consumption Distribution by Block

This pie chart shows the percentage contribution of each block to the total energy consumption.

- Block-3 contributes approximately **49.1%** of total energy
- EB Block-1 contributes around **40.8%**
- Admin Block contributes only about **10.1%**

The chart clearly indicates that nearly half of the campus energy is consumed by Block-3 alone. This highlights the need to focus energy-saving measures primarily on Block-3, followed by EB Block-1.

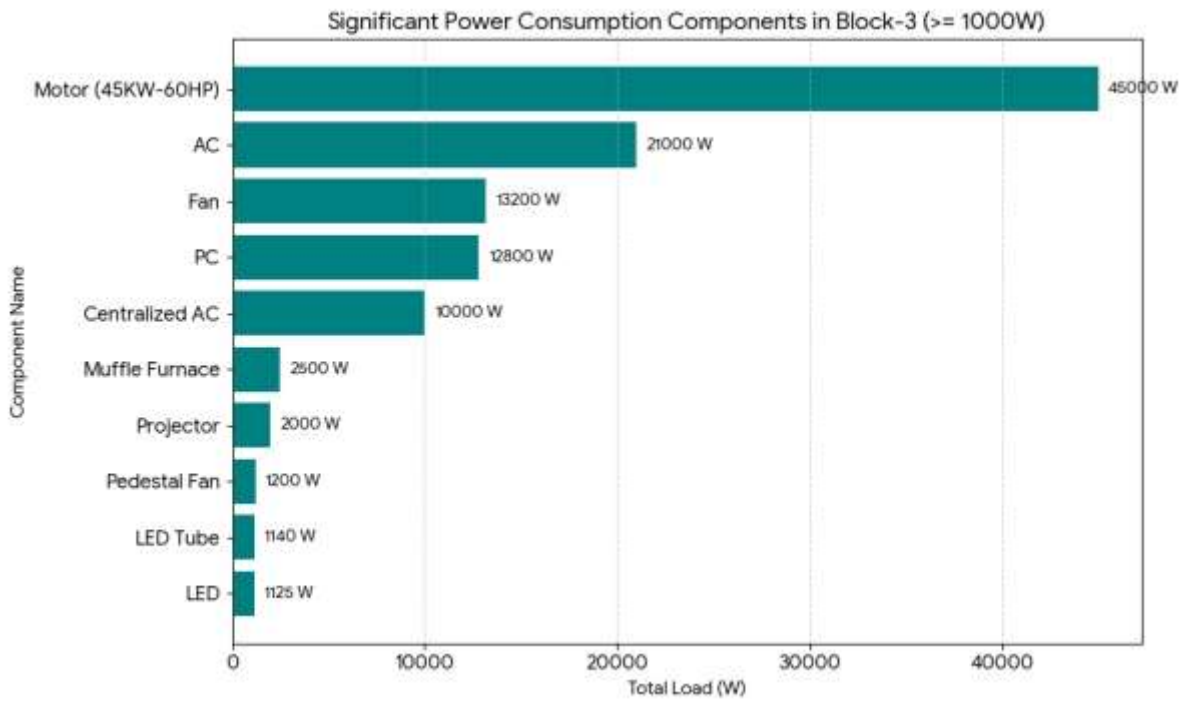


Figure 3 – Significant Power Consumption in Block-3

This bar chart displays major components in Block-3 that consume power greater than 1000 W.

- Motor (45kW–60HP) consumes the highest power at **45,000 W**
- AC units consume around **21,000 W**
- Fans and PCs contribute approximately **13,200 W** and **12,800 W** respectively
- Centralized AC adds another **10,000 W**

Other components such as projectors and furnaces consume relatively less power.

This indicates that **industrial motors and HVAC systems dominate energy consumption** in Block-3. These components are the primary targets for efficiency improvement.

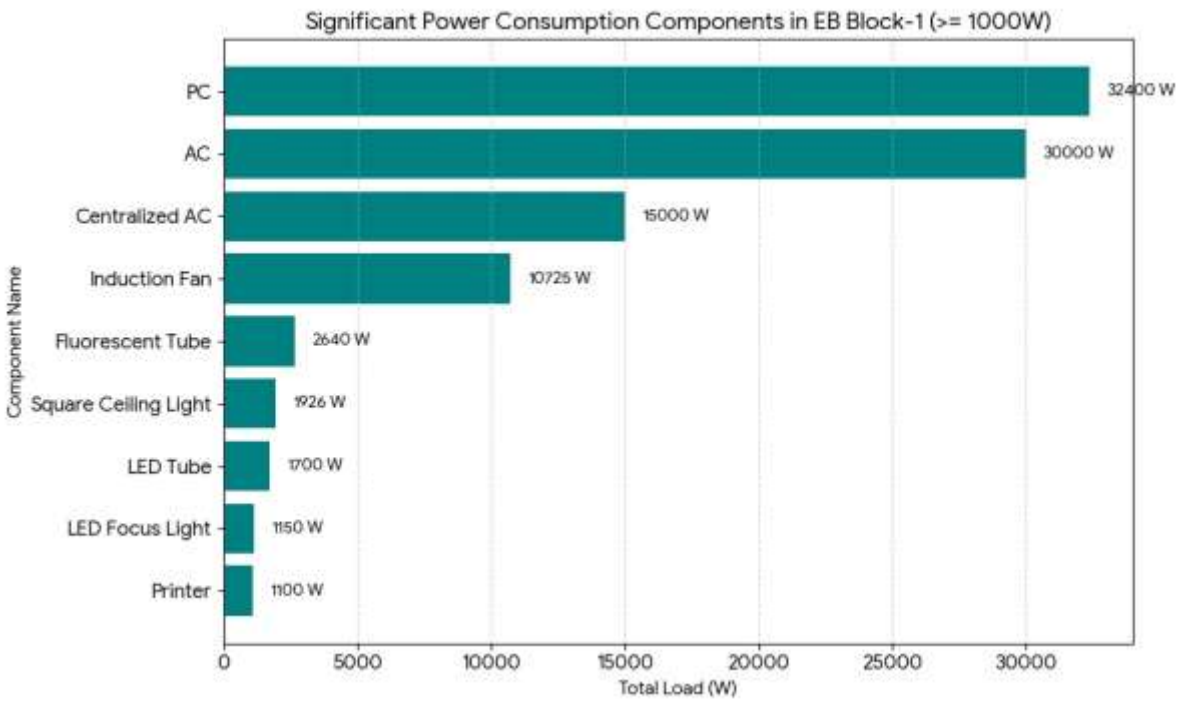


Figure 4 – Significant Power Consumption in EB Block-1

This chart shows major energy-consuming components in EB Block-1.

- PCs consume the highest power at **32,400 W**
- AC units follow with **30,000 W**
- Centralized AC contributes **15,000 W**
- Induction fans consume around **10,725 W**

Lighting and other components contribute comparatively less.

The results indicate that **plug loads (PCs) and AC systems are the major contributors** in this block. Efficient usage and control of these devices can reduce overall consumption.

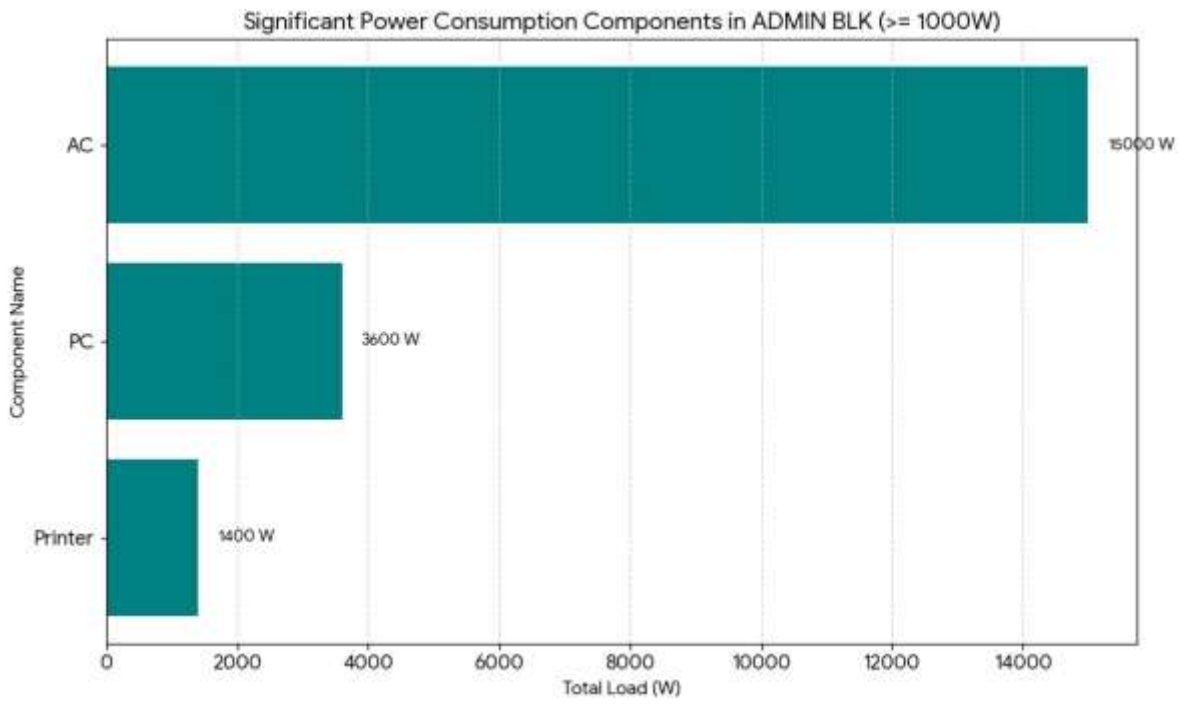


Figure 5 – Significant Power Consumption in Admin Block

This chart shows the major loads in the Admin Block.

- AC systems dominate with **15,000 W**
- PCs contribute **3,600 W**
- Printers consume around **1,400 W**

Compared to other blocks, Admin Block has fewer high-power components, resulting in lower total consumption.

The data shows that **AC systems are still the dominant load even in low-consumption blocks**, making them a key focus area.

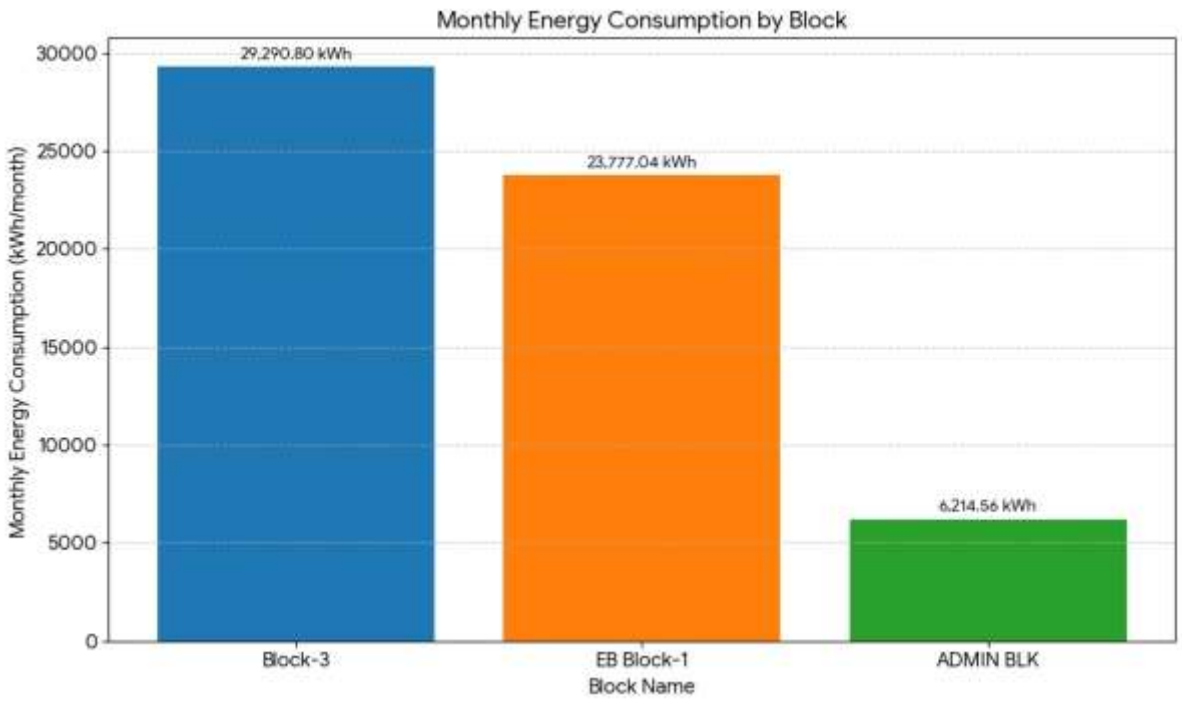


Figure 6 – Monthly Energy Consumption per Block

This bar chart represents the monthly energy consumption (in kWh/month).

- Block-3 consumes approximately **29,290.80 kWh/month**
- EB Block-1 consumes around **23,777.04 kWh/month**
- Admin Block consumes about **6,214.56 kWh/month**

The monthly data confirms the same trend observed in daily consumption. Block-3 consistently remains the highest energy consumer.

This analysis is important for:

- Estimating electricity bills
- Planning energy-saving strategies
- Identifying high-consumption zone

The diagrams show the energy consumption pattern across different campus blocks and components. Block-3 has the highest daily energy usage at about 976.36 kWh/day, followed by EB Block-1 at 792.57 kWh/day, while the Admin Block has the lowest consumption at around 207.15 kWh/day. The distribution chart confirms that Block-3 contributes nearly half of the total energy consumption (49.1%), with EB Block-1 contributing 40.8% and the Admin Block only 10.1%. Component-wise analysis indicates that in Block-3, high-power equipment such as motors (45,000 W) and air conditioners (21,000 W) dominate the load, along with contributions from fans and computers. In EB Block-1, computers and air conditioners are the major contributors, while in the Admin Block, air conditioning systems remain the primary load with relatively lower overall consumption. The monthly energy consumption data follows a similar trend, with Block-3 consuming the highest energy, followed by EB Block-1 and the Admin Block. Overall, the analysis shows that energy consumption is mainly influenced by high-power equipment, while other devices contribute based on their quantity and usage duration.

CHAPTER 7 – RESULTS AND DISCUSSION

7.1 Introduction

The results of the energy audit provide a clear understanding of the electrical energy consumption patterns across different campus blocks and floors. The collected data was analyzed based on component count, power rating, and distribution across various locations. This chapter presents the key findings and their interpretation to identify areas of high energy consumption and potential improvement.

7.2 Key Observations

1. Air Conditioning Systems Dominate Energy Consumption

The analysis shows that air conditioning (AC) systems are the highest energy-consuming components in all blocks. Even though the number of AC units is relatively lower compared to lighting and fans, their high power rating (typically 1–2 kW per unit) makes them the dominant load.

- AC systems contribute a major share of total load
- Centralized AC systems further increase consumption
- Continuous operation during working hours increases energy usage

This indicates that HVAC systems are the primary target for energy optimization.

2. Lighting Load is High Due to Quantity

Lighting systems such as LED tubes, fluorescent lamps, and square ceiling lights are present in large numbers across all floors.

- Individual power consumption is low (10–40 W)
- Total consumption becomes high due to large quantity
- Lighting is used for long durations (8–12 hours/day)

Even small improvements in lighting efficiency can result in significant energy savings.

3. Plug Loads from PCs and Equipment

Computers (PCs) and other electronic devices contribute significantly to the overall energy consumption.

- High number of PCs in labs and offices
- Continuous usage during working hours
- Standby losses also contribute

Plug loads are often overlooked but contribute a steady base load.

4. Floor-wise Consumption Pattern

The analysis shows that the **ground floor has the highest energy consumption**, followed by lower floors.

- Ground floor contains labs, offices, and high-usage areas
- Upper floors have comparatively fewer devices
- Equipment density decreases with height

This indicates that energy usage is highly dependent on functional space utilization.

5. Block-wise Energy Distribution

From the analysis:

- **Block-3** → Highest energy consumption
- **EB Block-1** → Moderate consumption
- **Admin Block** → Lowest consumption

This variation is due to:

- Number of rooms
- Type of activities
- Equipment density

7.3 Detailed Discussion

Effect of Number of Devices

Energy consumption increases with the number of devices, even if individual devices consume less power.

Example:

- One LED = low power
- Hundreds of LEDs = high total load

Quantity plays a crucial role in total energy usage.

Effect of Power Rating

High-power devices such as:

- AC units
- Motors
- Industrial equipment

consume more energy even in smaller numbers.

These loads significantly impact peak demand and electricity bills.

Effect of Usage Duration

Energy consumption is directly proportional to operating time.

- Lighting → used continuously
- AC → used during working hours
- PCs → used depending on activities

Longer usage leads to higher energy consumption.

Combined Impact

Total energy consumption depends on:

$$\textit{Energy} = \textit{Power} \times \textit{Time}$$

Therefore:

- High power + long duration = maximum energy usage
- Low power + high quantity = moderate consumption

7.4 Identification of Inefficiencies

The audit revealed several inefficiencies:

- Overuse of AC systems
- Unnecessary lighting during daylight
- Idle running of equipment
- Lack of energy monitoring systems

These inefficiencies lead to energy wastage.

7.5 Comparative Analysis

Parameter	High Impact	Medium Impact	Low Impact
AC Systems	✓		
Motors	✓		
PCs		✓	
Fans		✓	
Lighting			✓

This helps prioritize energy-saving measures.

7.6 Overall Findings

- Energy consumption is unevenly distributed
- Few devices contribute to most energy usage
- Many devices contribute through quantity
- Usage pattern strongly affects consumption

7.7 Summary

The results clearly indicate that energy consumption in the campus is influenced by multiple factors such as device count, power rating, and operational duration. High-power equipment like air conditioners and motors dominate the total load, while lighting and electronic devices contribute significantly due to their large numbers and continuous usage.

The findings provide a strong foundation for implementing energy-saving strategies and improving overall efficiency.

CHAPTER 8 – ENERGY CONSUMPTION ANALYSIS

8.1 Classification of Electrical Loads

Electrical loads are categorized based on their power rating, usage pattern, and contribution to total energy consumption.

8.1.1 Major Loads

Major loads contribute the highest share of total energy consumption due to high power ratings and extended operating duration.

Air Conditioners (AC)

Air conditioning systems are the dominant energy consumers.

- Typical power rating: 1–2 kW per unit
- Operate continuously during working hours
- High starting and running current

Impact:

Air conditioners significantly increase peak demand and overall energy consumption. Their usage directly affects electricity cost and load distribution.

Motors

Motors are used in pumps, ventilation systems, and laboratory equipment.

- Wide range of power ratings
- Continuous or periodic operation
- Efficiency varies with load and design

Impact:
Motors contribute substantially to energy consumption, especially when operating continuously or under inefficient conditions.

Computers (PCs)

Computers form a major portion of plug loads.

- Power rating: approximately 100–300 W
- Large number of units in laboratories and offices
- Frequent and long-duration usage

Impact:
Though individual consumption is moderate, the cumulative effect of multiple systems results in significant energy usage.

8.1.2 Medium Loads

Fans

Fans are used extensively for ventilation and comfort.

- Power rating: 60–100 W
- Large quantity across all blocks
- Operate for long durations

Impact:
Fans contribute moderately to total energy consumption. Their continuous usage increases their overall contribution.

8.1.3 Low Loads

LED Lighting Systems

LED lighting is widely used due to its energy efficiency.

- Power rating: 10–40 W
- High installation density
- Long operating hours

Impact:
Individual energy consumption is low; however, the total contribution becomes noticeable due to the large number of units.

8.2 Energy Distribution Analysis

The overall energy distribution indicates that:

- Major loads such as air conditioners, motors, and computers account for the largest share of total energy consumption
- Medium loads such as fans contribute a moderate portion
- Low loads such as LED lighting contribute less individually but collectively form a measurable portion of total energy

Energy consumption is therefore influenced by both power rating and quantity of devices.

8.3 Factors Affecting Energy Consumption

Power Rating

Higher power-rated devices consume more electrical energy per unit time and significantly influence total consumption.

Operating Duration

Energy consumption increases with longer usage duration. Equipment operating continuously contributes more to total energy usage.

Number of Devices

The total number of devices plays a crucial role. Even low-power devices can contribute significantly when used in large quantities.

Efficiency of Equipment

Inefficient equipment leads to higher energy losses and increased consumption.

8.4 Load Contribution Analysis

Load Category	Contribution Level	Characteristics
Air Conditioners	High	High power, long usage
Motors	High	Continuous operation, variable efficiency
Computers	High	Large quantity, steady usage
Fans	Medium	Moderate power, long usage
LED Lights	Low	Low power, high quantity

8.5 Discussion

Major loads dominate the total energy consumption due to their high power ratings. Air conditioning systems and motors are the primary contributors, especially in areas with continuous operation. Computers also contribute significantly due to their large number and consistent usage.

Medium loads such as fans contribute a steady portion of energy consumption because of their prolonged usage, despite having lower power ratings compared to major loads.

Low loads such as LED lighting consume less power individually but contribute to overall consumption due to their widespread use. This highlights the importance of considering both power rating and device quantity in energy analysis.

8.6 Insight

Even low-power devices become significant when used in large numbers. The cumulative effect of multiple low-power devices can result in considerable energy consumption, making it essential to optimize both high-power and low-power systems.

8.7 Summary

Energy consumption is determined by a combination of power rating, usage duration, and number of devices. High-power equipment contributes the largest share, while medium and low-power devices collectively influence total consumption. Proper classification of loads helps in identifying priority areas for improving energy efficiency and reducing overall energy usage.

CHAPTER 9 – RECOMMENDATIONS

9.1 Energy Saving Measures

Based on the analysis of energy consumption patterns across the campus blocks, several practical and cost-effective measures are recommended to improve energy efficiency and reduce overall electricity consumption.

1. Replacement of Conventional Lighting with LED Systems

Existing lighting systems such as fluorescent tubes and conventional bulbs should be replaced with LED lighting.

- LED lights consume 40–60% less energy
- Longer lifespan reduces maintenance cost
- Provides better illumination efficiency

Recommendation:
Gradual replacement of all existing lighting systems with LED fixtures across all blocks.

2. Use of Energy-Efficient Air Conditioning Systems

Air conditioners are the highest energy-consuming devices and require optimization.

- Replace old AC units with inverter-based AC systems
- Maintain temperature settings between 24–26°C
- Regular servicing improves efficiency

Recommendation:
Upgrade to high-efficiency AC systems and implement proper usage control to reduce unnecessary energy consumption.

3. Installation of Occupancy Sensors

Lighting and AC systems often remain ON even when rooms are unoccupied.

- Motion sensors automatically switch OFF devices when no occupancy is detected
- Reduces unnecessary energy wastage

Recommendation:
Install occupancy sensors in classrooms, laboratories, restrooms, and corridors.

4. Use of Energy-Efficient Motors

Motors used in various applications should be optimized for efficiency.

- Replace old motors with high-efficiency motors (IE2/IE3 standards)
- Operate motors near rated load conditions
- Avoid frequent starting and stopping

Recommendation:
Adopt energy-efficient motors and ensure proper maintenance to reduce losses.

5. Scheduling and Load Management

Unplanned usage of electrical equipment leads to energy wastage.

- Switch OFF equipment when not in use
- Avoid simultaneous operation of heavy loads
- Optimize operating schedules

Recommendation:
Implement structured schedules for operating equipment in laboratories and offices.

6. Implementation of Smart Energy Monitoring Systems

Real-time monitoring helps in tracking and controlling energy consumption.

- Use smart meters and IoT-based systems
- Monitor block-wise and floor-wise energy usage
- Identify abnormal consumption patterns

Recommendation:
Install a centralized energy monitoring system for continuous tracking and analysis.

9.2 Additional Recommendations

Awareness and Behavioral Changes

- Educate users about energy conservation
- Encourage switching off unused equipment
- Promote energy-saving practices

Regular Maintenance

- Clean and service AC systems periodically
- Check wiring and connections
- Replace faulty equipment

Power Factor Improvement

- Install capacitors where required
- Reduce reactive power losses
- Improve overall system efficiency

Renewable Energy Integration

- Install solar panels on rooftops
- Utilize renewable energy sources
- Reduce dependency on grid supply

9.3 Expected Benefits

Implementation of the above measures can result in:

- Reduction in energy consumption by 20–30%
- Lower electricity bills
- Improved equipment efficiency
- Increased system reliability
- Reduced environmental impact

9.4 Summary

The recommended measures focus on improving efficiency in major, medium, and low electrical loads. By combining technological improvements with proper management practices, significant energy savings can be achieved. These recommendations provide a practical roadmap for enhancing energy efficiency and promoting sustainable energy usage within the campus.

CHAPTER 10 – CONCLUSION

The energy audit carried out for the selected campus blocks provided a clear and systematic understanding of the energy consumption patterns and distribution of electrical loads. Through detailed data collection and analysis, the study successfully identified the major energy-consuming equipment such as air conditioning systems, motors, and computers, along with supporting loads like lighting and fans.The analysis revealed that energy consumption is influenced by multiple factors including the number of devices, their power ratings, and the duration of operation. High-power equipment such as air conditioners and motors contribute significantly to the overall load, while low-power devices like LED lights also play an important role due to their large numbers and continuous usage. Floor-wise and block-wise evaluations highlighted that areas with higher equipment density and usage, particularly ground floors and specific blocks, consume more energy.The audit also identified several inefficiencies such as unnecessary operation of equipment, lack of proper monitoring systems, and the use of outdated or less efficient devices. These issues contribute to increased energy consumption and operational costs.The implementation of the recommended energy-saving measures, including the use of LED lighting, energy-efficient air conditioning systems, optimized motor usage, occupancy-based control systems, and smart energy monitoring, can significantly improve the overall efficiency of the system. Proper scheduling and regular maintenance

further enhance the effectiveness of these measures.By adopting these improvements, the institution can achieve a considerable reduction in energy consumption, leading to lower electricity costs and improved operational efficiency. In addition, these measures support environmental sustainability by reducing energy wastage and minimizing carbon emissions.Overall, the energy audit serves as an effective tool for identifying inefficiencies and providing practical solutions for energy management. The findings of this study can be used as a foundation for implementing long-term energy conservation strategies and promoting responsible energy usage within the campus.

APPENDICES

Appendices are included to provide supplementary information that supports the analysis and findings presented in the main body of the report. These details enhance the clarity and reliability of the study without interrupting the flow of the core content.

APPENDIX 1 – RAW DATA

The raw data collected during the energy audit includes detailed information on the number and types of electrical components present in each block and floor. This data was gathered through direct observation and manual counting.

Details Included:

- Number of lighting units (LED tubes, square lights, etc.)
- Number of fans
- Number of air conditioning units
- Number of computers (PCs)

- Number of motors and other electrical equipment
- Block-wise and floor-wise distribution of components

This dataset forms the basis for all calculations and analysis carried out in the project. It helps in identifying load distribution and high-consumption areas.

APPENDIX 2 – SURVEY FORMS

Survey forms were used during the field investigation phase to systematically record observations from each location.

Contents of Survey Forms:

- Block name and floor details
- Room type (classroom, lab, office, etc.)
- List of electrical equipment present
- Quantity of each component
- Estimated usage duration
- Observations on equipment condition

These forms ensured uniform data collection and helped maintain accuracy and consistency across different locations.

APPENDIX 3 – CALCULATIONS

The calculations performed in this study are based on standard electrical energy formulas.

1. Power Calculation

$$Power\ (W) = Voltage \times Current$$

2. Energy Consumption

$$Energy\ (kWh) = Power\ (kW) \times Time\ (hours)$$

3. Total Load Calculation

$$Total\ Load = \sum (Number\ of\ Devices \times Power\ Rating)$$

4. Efficiency Consideration (for Motors)

$$Efficiency = Input\ Power\ /\ Output\ Power \times 100$$

5. Daily Energy Usage

$$Energy = Total\ Load \times Operating\ Hours$$

6. Monthly Energy Consumption

$$Monthly\ Energy = Daily\ Energy \times Number\ of\ Days$$

These calculations were used to estimate the energy consumption of different components and to compare their contribution to the overall load.

SUMMARY OF APPENDICES

The appendices provide supporting data, structured survey formats, and detailed calculations that validate the results and conclusions of the energy audit. They serve as a reference for further analysis and future energy studies.

REFERENCES

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2. Barnard, R.W. (1980) Electrical Analysis
3. Shin, K.G. (1984) Control Systems